Comparison of Thumb Models using Cadaveric Joint Angle Data

Paolo Bosques-Paulet[1,2], Joseph Towles[1]
Department of Engineering[1], Department of Mathematics[2]
Swarthmore College

IEEE EMBC 2026 48TH ANNUAL International conference of the IEEE Engineering in Medicine and Biology Society | 26-30 July 2026 Toronto, Canada

Poster No.

IEEE EMBS IEEE Engineering Medicine & Biology Society

Motivation

Restoring stable grasp in people with neurological or musculoskeletal impairments requires understanding thumb kinematics. The kinematic structure of the thumb, necessary for movement and force production, is a matter of debate. Previous studies have found and employed different kinematic structures [1][2] and previous studies have pointed to the importance of including a translational DOF at the thumb's base joint given observable motion of the first metacarpal bone toward the wrist when individual thumb muscles are loaded in situ. [3] This study used mathematical modeling and cadaveric experiments to examine how different thumb kinematic structures influence extrinsic muscle (FPL, APL, EPL,EPB) endpoint displacements.

Methods: Forward Kinematic Thumb Models

The following are both systems of links and joints

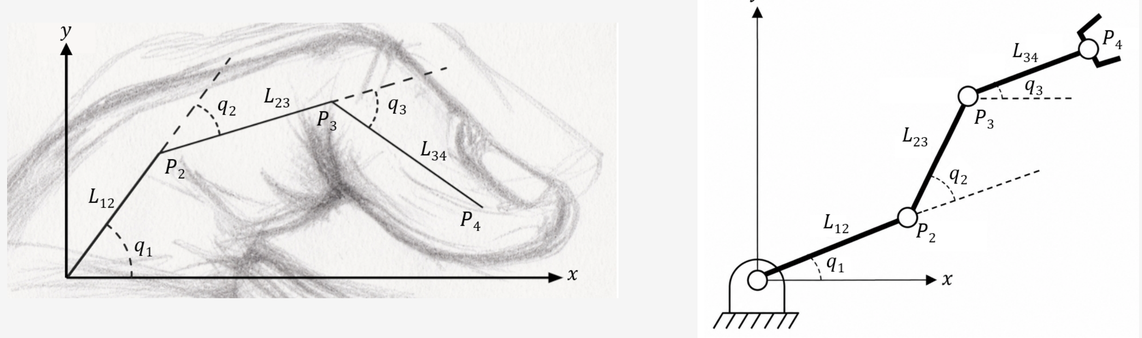


Figure 1: Kinematic Structures

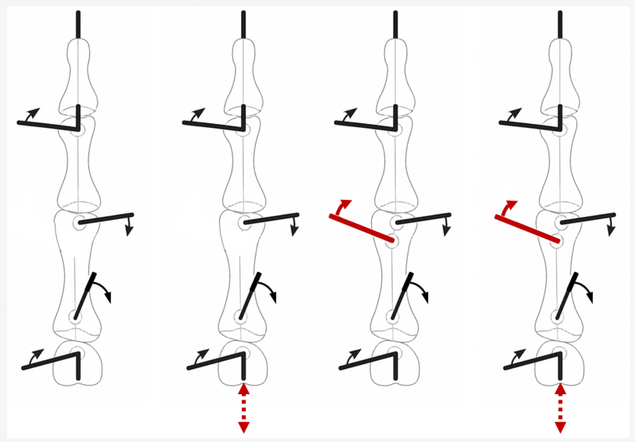
In this work we borrow Forward Kinematics (FK) from robotics. We utilize the Denavit-Hartenburg (DH) convention of FK for our 3D models.[4] The general application of these techniques is presented.

$$L, q \xrightarrow{FK} P \qquad \theta, d, a, \alpha \xrightarrow{DH} P$$

- L Link lengths
- q Joint angles
- P Joint positions
- θ Joint angles
- d Link offsets
- a Link lengths
- α Joint twists

Methods: Structural Variation

In this study we compare four distinct model formulations ranging from 4 to 6 degrees of freedom, as supported by the literature.



Additional Joint Types:

← Ad/abduction Axis

← Translation Joint

Figure 2: Model Structures (4DOF, 5DOF+Trans, 5DOF+AA, 6DOF)

Methods: Actuating Models

Joint-angle/% muscle-exertion pairs from [5] were combined with kinematic models adapted from [6] to predict thumb endpoint trajectories.

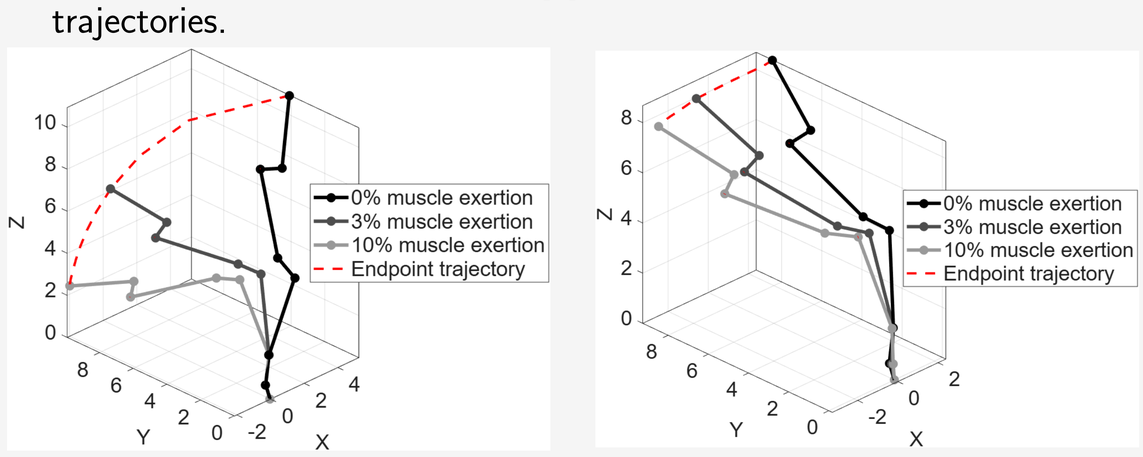


Figure 3: FPL and APL Example Trajectories using Structure 3 (5DOF+AA) (Figure)

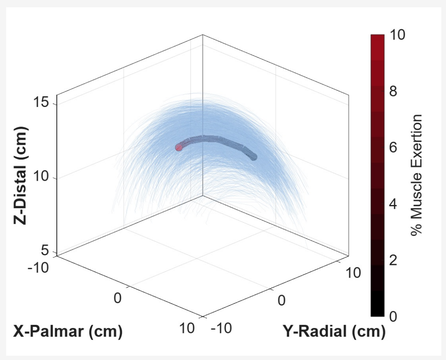
Methods: Individual Trial Models

Instances of each structure were generated by Monte Carlo sampling from the parameter distributions for model 3 in [6]. The magnitude of the change in the distal phalanx vector was used to assess convergence as new instances of a model were added. To compare thumb endpoint trajectories, in each of the four models, endpoints were translated to a common reference frame as shown in figures 4(a) and 4(b). A 95% confidence ellipsoid for the endpoint distribution was then computed for each model using the χ^2 distribution with 3 degrees of freedom as shown in figures 4(c) and 4(d). Overlap was computed by sampling 500000 points and determining which ellipsoid they resided in to estimate ellipsoid volume and intersections between each structure. For structures S_1 and S_2,

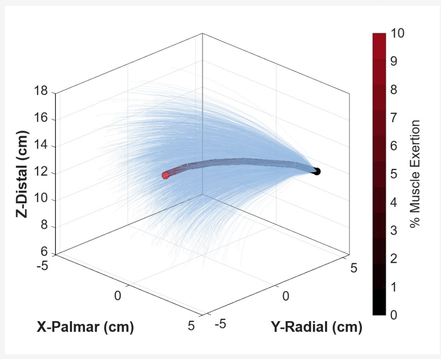
$$\text{Jaccard}(S_1,S_2) = \frac{S_1 \cap S_2}{S_1 \cup S_2}, \quad \text{Rel. Overlap}(S_1, S_2) = \frac{S_1 \cap S_2}{\min(S_1, S_2)}.$$

Results

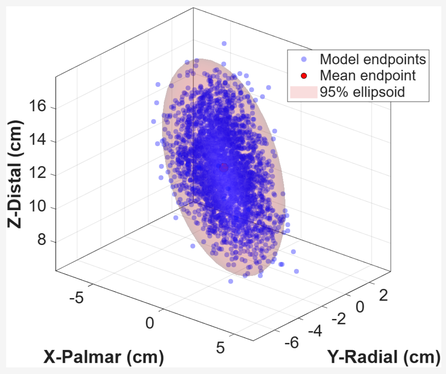
Below are results for the EPL muscle (Results were generated for FPL, APL, and EPB as well)



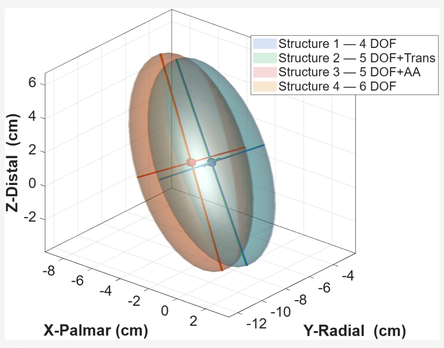
(a) EPL Endpoint Trajectories



(b) EPL Endpoint Trajectories from Common Reference



(c) EPL 95% Endpoint Confidence Ellipsoid



(d) Overlap of 4 Different Model Structures

Figure 4: Summary Results - (a)-(c) using Structure 3

EPL Model Pair	Intersection (cm^3)	Jaccard	Rel. Overlap
Structure 1 vs Structure 2	152.6151	0.967	0.984
Structure 1 vs Structure 3	116.4905	0.537	0.750
Structure 1 vs Structure 4	113.8244	0.518	0.733
Structure 2 vs Structure 3	118.9313	0.555	0.767
Structure 2 vs Structure 4	116.2564	0.536	0.750
Structure 3 vs Structure 4	175.3634	0.968	0.984

Discussion

Including an adduction–abduction (AA) degree of freedom at the metacarpophalangeal (MCP) joint produced substantial shifts in endpoint distribution across all extrinsic thumb muscles excluding the FPL. This suggests that cadaveric motion data may be sufficient to determine whether the MCP-AA joint is physiologically necessary in kinematic thumb models. In contrast, adding a translational degree of freedom at the carpometacarpal (CMC) joint had a much smaller effect. However, these results are based on actuator inputs spanning only 10% of the theoretical muscle-exertion range. Greater separation between models may emerge when a broader range of muscle activation is considered.

References

- D. C. McFarland, B. I. Binder-Markey, J. A. Nichols, S. J. Wohlman, M. de Bruin, and W. M. Murray, "A musculoskeletal model of the hand and wrist capable of simulating functional tasks," *IEEE Transactions on Biomedical Engineering*, vol. 70, no. 5, pp. 1424–1435, 2023.
- F. J. Valero-Cuevas, M. E. Johanson, and J. D. Towles, "Towards a realistic biomechanical model of the thumb: The choice of kinematic description may be more critical than the solution method or the variability/uncertainty of musculoskeletal parameters," *Journal of Biomechanics*, vol. 36, pp. 1019–1030, July 2003.
- J. D. Towles, V. R. Hentz, and W. M. Murray, "Use of intrinsic thumb muscles may help to improve lateral pinch function restored by tendon transfer," *Clinical Biomechanics*, vol. 23, pp. 387–394, May 2008.
- J. Denavit and R. S. Hartenberg, "A kinematic notation for lower-pair mechanisms based on matrices," *Journal of Applied Mechanics*, vol. 22, no. 2, pp. 215–221, 1955.
- Z.-M. Li, J. Tang, M. Chakan, and R. Kaz, "Complex, multidimensional thumb movements generated by individual extrinsic muscles," *Journal of Orthopaedic Research*, vol. 26, pp. 1289–1295, Sept. 2008.
- V. J. Santos and F. J. Valero-Cuevas, "Reported anatomical variability naturally leads to multimodal distributions of denavit–hartenberg parameters for the human thumb," *IEEE Transactions on Biomedical Engineering*, vol. 53, pp. 155–163, Feb. 2006.

SWARTHMORE | ENGINEERING NEUROMUSCULAR BIOMECHANICS LAB